



IIT Gandhinagar

PG Diploma in GenAI-Powered Data Science & Engineering

Curriculum and Learning Journey





Credit Structure (40 Credits)

Component	Credits	Hours
Theory (Lectures)	17	170 hrs
Practical (Labs + Capstone)	16	320 hrs
Competitive Coding & Ideas Lab	7	140 hrs
TOTAL	40	630 hrs

Credit Calculation: 10 teaching hours = 1 credit OR 20 practical hours = 1 credit

20-Week Learning Journey

Phase	Timeline	Subjects & Topics	Job Skills Unlocked
Foundation	Weeks 1-6	SQL & Data Modeling, Python for DS, Statistics, ML Fundamentals	Data manipulation, EDA, statistical analysis, ML model training
Specialization	Weeks 7-14	Advanced ML & Deep Learning, Time Series, BI & Visualization, GenAI for DS	Neural networks, NLP, forecasting, dashboards, text-to-SQL, automated insights
Midterm	Week 15	Comprehensive technical assessment	Mid-program evaluation, interview readiness check
Capstone	Weeks 16-19	Complete ML system: data pipeline, models, API, dashboards, GenAI analytics	End-to-end ML pipelines, model deployment, A/B testing, stakeholder presentations
Final Exam	Between Week 15-20	SQL mastery, ML coding, case studies, system design	Interview-ready for Data Scientist, ML Engineer, Analytics Engineer roles



Month-by-Month: Subjects, Tools & Job Skills

Month 1 (Weeks 1-6): Engineering Toolkit

- Subjects: SQL & Data Modeling (6cr), Python for DS (4cr), Statistics (5cr), ML Fundamentals (6cr)
- Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, Jupyter, scikit-learn, Git
- Job Skills: Data manipulation, EDA, statistical analysis, ML model training (regression, classification, clustering)

Month 2 (Weeks 7-10): DS Specialization

- Subjects: Advanced ML & Deep Learning (7cr), Time Series (4cr), BI & Visualization (5cr), GenAI for DS (3cr)
- Tools: TensorFlow/PyTorch, XGBoost, Prophet, ARIMA, Tableau, Power BI, Plotly, LangChain, OpenAI API
- Job Skills: Neural networks, computer vision, NLP, forecasting, interactive dashboards, text-to-SQL, automated insights

Months 3-4 (Weeks 11-17): Capstone Project

- Capstone (9cr): Complete ML system with data pipeline, models, API, dashboards, GenAI-powered analytics
- Tools: Full DS stack, MLflow (experiment tracking), Docker, FastAPI, Streamlit, AWS (S3, SageMaker)
- Job Skills: End-to-end ML pipelines, model deployment, A/B testing, business metrics, stakeholder presentations

Month 5 (Weeks 18-20): Interview Mastery

- Focus: 200+ SQL problems, ML coding, case study practice, system design for ML, 15+ mock interviews
- Tools: LeetCode SQL, HackerRank, ML interview platforms, case study frameworks
- Job Skills: SQL mastery, ML theory deep-dives, business problem solving, metrics design, statistical significance

Throughout 20 Weeks: Competitive Coding & Ideas Lab (7cr)

- Daily SQL practice, Python DSA, Kaggle competitions, data ethics, product thinking for data products



Extra Value Pack

Data Science Tools Mastery

- Python (Pandas, NumPy), SQL (PostgreSQL, MySQL), scikit-learn, XGBoost, visualization (Matplotlib, Seaborn, Plotly)

GenAI for Data Science

- LangChain, OpenAI/Claude APIs, automated insights, text-to-SQL, natural language analytics, prompt engineering for data

BI & Analytics Platforms

- Tableau, Power BI, Looker, Streamlit, dashboard design, storytelling with data, executive presentations

ML Production Stack

- MLflow, Docker, FastAPI, AWS/GCP, model deployment, experiment tracking, A/B testing, feature stores

Agile for Data Teams

- Sprint planning for analytics, Jira for data projects, stakeholder management, data quality frameworks

Professional Development

- 2 clubs: Entrepreneurs Club + AI Innovation Club (optional, Mon/Wed/Fri evenings). Kaggle competitions & case study discussions



Daily Learning Structure

Time (24-hour)	Activity
07:45 - 09:15	Breakfast at IIT-GN Campus
09:15 - 10:45	Session 1A - Core Concepts: Instructor teaches ML/DS fundamentals, SQL, statistics with live demos and Q&A
11:00 - 12:45	Session 1B - Hands-On Labs: Data wrangling, model building, dashboard creation. Pair programming, live debugging
12:45 - 14:00	Lunch at IIT-GN Campus + Squad Sync: Discuss data insights, share analysis wins, plan afternoon work
14:00 - 15:15	Session 2A - Production Patterns: Real-world DS projects, GenAI analytics, business case studies, industry practices
15:30 - 17:00	Session 2B - Advanced Implementation: ML pipelines, feature engineering, mini-projects, peer reviews
17:00 - 17:30	Break + Snacks at IIT-GN Campus
17:30 - 18:00	Daily Standup: What insights were found? What broke? Blockers? 2-min data story presentations. Commit to GitHub
18:00 - 19:30	SQL & Python Coding Hour: Complex queries, DSA practice, Kaggle competitions, daily leaderboard
19:30 onwards	Dinner at IIT-GN Campus, then hostel
Saturday	Weekly assessment + case study presentations + mock DS interviews + Kaggle competitions + club activities (optional)

Note: All meals (breakfast, lunch, snacks, dinner) are at IIT Gandhinagar campus mess